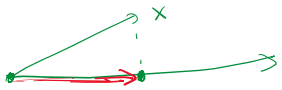
Lecture 21

Wednesday, April 1, 2026 4:02 PM

Common issues in quiz 4/HW 8

1. Projection of x to the direction of w in quiz 4:
2. Proof in problem 3.1 in HW8:
   Make sure that the solution is the *smallest* eigenvalue,
   By checking the objective value at all critical points
3. Figures in Problem 4 in HW8-- please read instruction

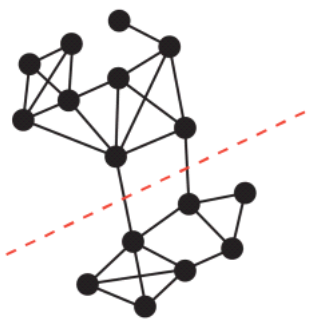


$\text{proj}_w x = \frac{x \cdot w}{w \cdot w} w$, $\|\text{proj}_w x\| = \frac{|x \cdot w|}{\|w\|}$

$\|\text{proj}_w x\| = x \cdot w$ only when $\|w\| = 1$

Today: (Section 9.4, binary spectral clustering)

Min-cut problem: find a partition so that there are few edges connecting the two components



Examples of similarity graph: two points connected if they are "similar".

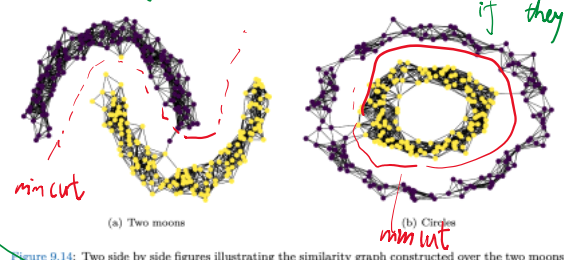


Figure 9.14: Two side by side figures illustrating the similarity graph constructed over the two moons and circles data sets.

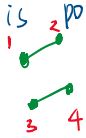
min cuts leads to clustering ...

Graph Laplacian and its associated properties:

$W \in \mathbb{R}^{m \times m}$: weight matrix, $D = \begin{bmatrix} d_1 & & & \\ & d_2 & & \\ & & \ddots & \\ & & & d_m \end{bmatrix}$ diagonal. $d_i = \sum_{j=1}^{m} w_{ij}$ (sum of i-th row of W)

Graph Laplacian : $L = D - W$

property : L is positive semi-definite, has an eigenvalue of ~~zero~~ 0 with eigenvector of $\begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$

Example. Four nodes 

Two edges (1, 2) (3, 4)

then $W = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ $D = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$ $L = D - W = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}$

eigenvalues : 0, 0, 2, 2

eigenvectors $\frac{1}{2}\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, $\frac{1}{2}\begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$, $\frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}$, $\frac{1}{\sqrt{2}}\begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}$

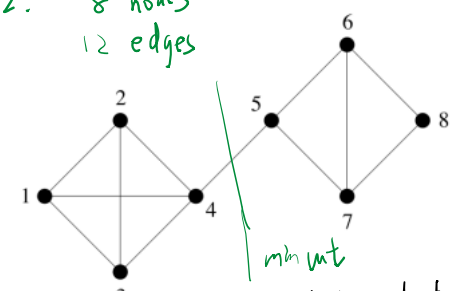
Note. min cut has a natural solution of group 1 : (1, 2) group 2 (3, 4), which corresponds to one eigenvector.

Definition 9.23 The second smallest eigenvalue/eigenvector is called Fiedler eigenvalue eigenvector. denoted by λ_F.

Theorem 9.25 : $\lambda_F = \min \{ x^T L x \mid \|x\| = 1, \mathbb{1}^T x = 0 \}$

vector of ones

Example 2. 8 nodes
12 edges



8 eigenvalues of L: 0, 0.3, 2.3, 4, 4, 4, 4, 5.3

λ_F : fiedler eigenvalue is relatively small, indicates that the graph is not especially well-connected.

eigenvectors $\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$, $\begin{bmatrix} \\ \\ \\ \end{bmatrix}$

Section 9.4 Binary spectral clustering.

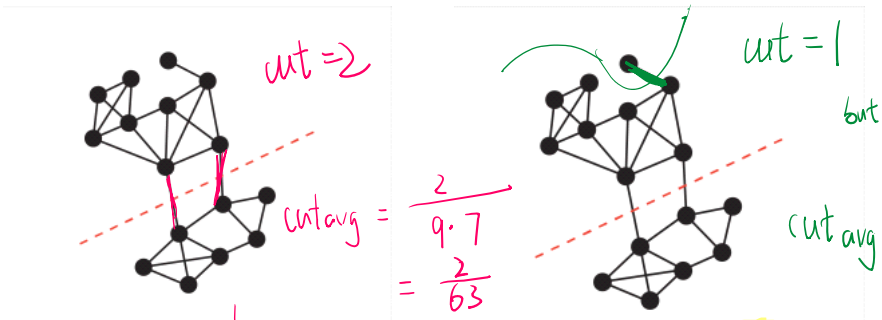
Section 9.4 Binary spectral clustering

min cut: find a partition so that there are few edges connecting the two components

Math formulation: partition set of nodes N into A and $A^c = N \setminus A$

so that $\text{cut}(A) = \sum_{i \in A} \sum_{j \in A^c} w_{ij}$ is minimized.

- However, this might choose a set where A is small or A^c is small. for example, containing a single point.



cut = 2

$\text{cut}_{avg} = \frac{2}{9 \cdot 7} = \frac{2}{63}$

cut = 1 is better that cut = 2 but A would have only one node...

$\text{cut}_{avg} = \frac{\text{cut}}{1 \cdot 15} = \frac{1}{15}$

To encourage that the clusters have similar sizes, we can normalize such as $\text{cut}_{avg}(A) = \frac{\text{cut}(A)}{\# A \cdot \# A^c}$

this cut is preferred using cut_{avg}

ensure neither A nor A^c is small

Next, the goal is to minimize . In fact, it is NP-hard.

Instead, solve a "similar" problem:

spectral clustering procedure

1. set $b = \frac{\# A}{\# A^c}$, $u = \mathbb{1}_A - b\,\mathbb{1}_{A^c}$

   indicator vector. has values of 0, 1 $\begin{cases} 1 & \text{if the node is in } A \\ 0 & \text{otherwise} \end{cases}$

   Then $\mathbb{1}^T u = 0$, and

   $\text{cut}_{avg}(A) = \frac{1}{m} \frac{u^T L u}{\|u\|^2}$

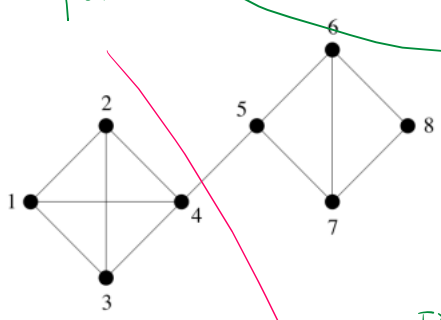
2. minimize instead: The solution is the fiedler eigenvector of L (the second smallest eigenvector)

   solve $\min \left\{ \frac{u^T L u}{\|u\|^2} : \mathbb{1}^T u = 0 \right\}$

3. Denote the fiedler eigenvector by u_F, then we identify A by the signs of u_F: $A = \{ i : [u_F]_i > 0 \}$, $A^c = \{ i : [u_F]_i \leq 0 \}$

Theorem 9.31

Example.



Graph Laplacian.

$$L = D - W = \begin{pmatrix} 3 & -1 & -1 & -1 & 0 & 0 & 0 & 0 \\ -1 & 3 & -1 & -1 & 0 & 0 & 0 & 0 \\ -1 & -1 & 3 & -1 & 0 & 0 & 0 & 0 \\ -1 & -1 & -1 & 4 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 3 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 3 & -1 & -1 \\ 0 & 0 & 0 & 0 & -1 & -1 & 3 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & -1 & 2 \end{pmatrix}.$$

Fiedler eigenvector:

$$\mathbf{v}_2 = \begin{pmatrix} 0.3747 \\ 0.3747 \\ 0.3747 \\ 0.2553 \\ -0.1843 \\ -0.3747 \\ -0.3747 \\ -0.4458 \end{pmatrix}$$

positive: so nodes 1, 2, 3, 4 is in a group

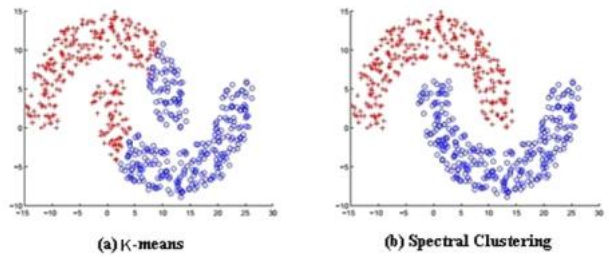
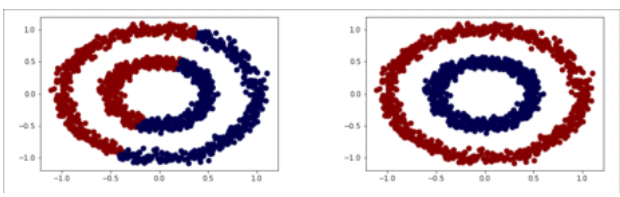
negative: nodes 5, 6, 7 & 8 is in another

solution

$$\begin{pmatrix} -0.1845 \\ -0.3747 \\ -0.3747 \\ -0.4458 \end{pmatrix}$$ } negative

nodes 5, 6, 7, 8 is in another group.

Comment: spectral clustering can succeed when k-means fail:



(a) K-means (b) Spectral Clustering